

SketchGuide+: Enabling Parametric Freehand Robot Programming via Wearable-Assisted Sketching

Christoph Heindl*
cheind@profactor.at

Visual Computing, Profactor GmbH
Steyr, Austria

Gerhard Ebenhofer
gebenh@profactor.at

Visual Computing, Profactor GmbH
Steyr, Austria

Sezgin Kutlu
skutlu@profactor.at

Visual Computing, Profactor GmbH
Steyr, Austria

Fabian Widmoser
fwidmo@profactor.at
Robotics, Profactor GmbH
Steyr, Austria

Andreas Pichler
apichl@profactor.at
Robotics, Profactor GmbH
Steyr, Austria

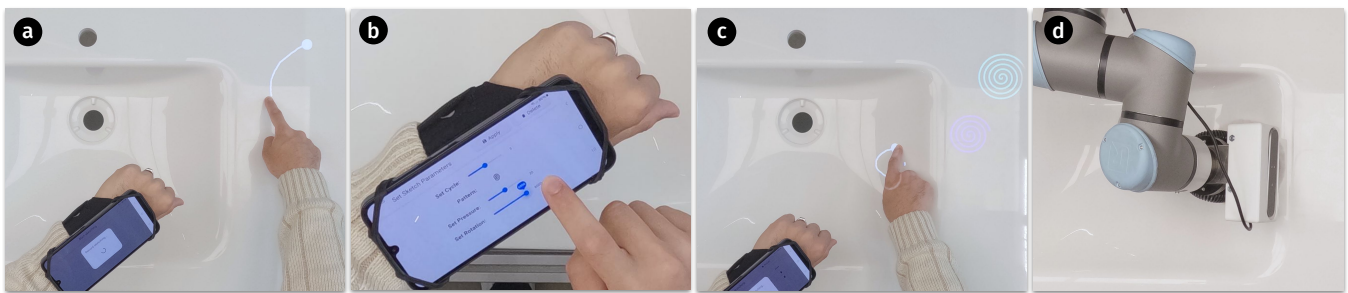


Figure 1: *SketchGuide+* enables expressive, model-free robot programming by combining freehand sketching on physical surfaces with task-level parameter input via a smartwatch-style wearable. Our system operates using a single RGB-D camera and a wearable device, without requiring CAD models or tracked tools - footnote 1 has a video link. (a) A user marks the area to be processed directly on the workpiece using fingertip sketches. While sketching, immediate system feedback is provided by projecting the recognized strokes onto the surface. (b) For the active sketch, task-specific parameters are adjusted on the wearable using the drawing hand. (c) The system supports creating multiple sketches per session, each enriched with task-specific parameters and aligned to the surface geometry. (d) Captured sketches and parameters are converted into task-aware end-effector trajectories and executed based on the robot's kinematics and surface geometry.

Abstract

We present *SketchGuide+*, an extension of SketchGuide [6] for intuitive robot programming through freehand sketching on real-world surfaces. Building on the original system's ability to capture fingertip sketches on arbitrary geometries using a single RGB-D camera, *SketchGuide+* incorporates a smartwatch-style wearable for real-time parameter control and a projector that delivers immediate on-surface visual feedback, increasing user immersion during the sketching process. The system temporally aligns wearable input with sketch trajectories to produce parameter-rich robot motion plans. We demonstrate the system in a basin polishing scenario

and conduct a preliminary user study comparing *SketchGuide+* to a baseline without projection or wearable input. Results indicate improved user confidence, smoother interaction flow, and reduced task completion time. An accompanying video ¹ illustrates the end-to-end interaction and robot execution workflow. These findings underscore the potential of combining vision, wearables, and spatial projection to enable expressive, low-friction robot programming accessible to non-experts.

CCS Concepts

• Human-centered computing → Mixed / augmented reality; • Computing methodologies → Machine learning approaches; Vision for robotics.

Keywords

Human Machine Interaction; Computer Vision; Spatial Augmentation; Machine Learning

ACM Reference Format:

Christoph Heindl, Gerhard Ebenhofer, Sezgin Kutlu, Fabian Widmoser, and Andreas Pichler. 2025. SketchGuide+: Enabling Parametric Freehand

*Corresponding author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '25, Espoo, Finland.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1477-1/25/10

<https://doi.org/10.1145/3714394.3754389>

¹ Additional material <https://github.com/cheind/SketchGuidePlus>

Robot Programming via Wearable-Assisted Sketching. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754389>

1 Introduction and Related Work

Robots play a critical role in modern manufacturing, enabling increased efficiency, precision, and flexibility. However, programming robots for diverse and evolving tasks remains a bottleneck, especially for non-expert users operating in low-volume or custom production contexts.

While recent advances have explored multi-modal interfaces and wearable sensors [1, 5, 13], these solutions often rely on complex hardware setups or constrained interaction paradigms. Several systems rely on tracked or handheld devices [3, 12] to specify robot trajectories with high spatial fidelity. Others use stylus-based or touch-sensitive surfaces [4, 10], often tied to specialized hardware or structured workspaces. Gesture-based methods [7, 14] enable tele-operation via hand or arm poses but typically lack the spatial precision required for direct, surface-constrained interaction. More recent work explores natural interfaces for robot programming, such as pen-and-voice systems [11], augmented reality for in-situ teaching [2, 9], and wearable AR programming tools [8], yet these still depend on headsets, tracked controllers, or haptic gloves.

In contrast, freehand sketching on physical workpieces offers a direct, intuitive means of specifying approximate geometric intent—well suited to tasks such as polishing, painting, or inspection. Yet remains underexplored in human-robot collaboration (HRC) and task-level robot programming. Building upon this insight, SketchGuide [6] introduced a baseline system that leverages a single RGB-D camera to capture 3D fingertip sketches on arbitrary surfaces, enabling model-free, real-time programming through estimated geometric contact.

2 Contribution

In this work, we extend this concept through *SketchGuide+*, a multi-modal, vision-based system that enhances expressivity and user control by integrating two key contributions:

- **Wearable Interface.** A smartwatch-style wearable worn on the forearm, which allows users to set task-relevant sketch parameters (e.g., operation type, force, or intent) in real time using the same finger used for sketching.
- **Reactive Projection.** A projector-based feedback system that provides immediate on-surface visualization of the interpreted sketch and parameters, increasing immersion and task awareness.

Unlike systems that require CAD models, tracked tools, or offline programming environments, *SketchGuide+* supports in-situ programming on arbitrary surfaces using uninstrumented hands and workpieces (see Figure 1). The input channel of the wearable is tightly synchronized with the vision pipeline, enabling real-time parameter annotations to be seamlessly fused with sketch trajectories. These are then translated into parameter-rich robot motion plans for downstream execution. We demonstrate *SketchGuide+* in a basin polishing task, where users must not only define approximate polishing regions but also specify segment-level parameters.

3 Approach

Building on Figure 2 we provide the implementation details of *SketchGuide+*.

3.1 SketchGuide

For completeness, we briefly summarize SketchGuide [6], which forms the foundation of our work. SketchGuide employs a single RGB-D camera to jointly estimate 3D hand pose and reconstruct the geometry of the workpiece surface in real time. Hand keypoints are detected in the RGB image using a monocular pose estimator and are subsequently lifted to 3D by fusing depth from the RGB-D sensor with predictions from a monocular depth model to compensate for noise and occlusions. The system detects oriented contact points by identifying when the fingertip is sufficiently close to the estimated surface geometry and aligned with local surface normals. These contact events are further segmented over time into coherent 3D sketch trajectories based on hand gestures and temporal patterns. Each trajectory consists of a sequence of surface-projected points enriched with local geometric context, making them suitable for downstream applications such as robot path planning, task execution, and simulation. The design is model-free, requiring no prior knowledge of the workpiece, and supports sketching on arbitrarily shaped objects, making it particularly well suited for small-batch or flexible manufacturing tasks.

3.2 Wearable Interface

The wearable connects to *SketchGuide+* over Wi-Fi and runs a web-based application that enables real-time, bi-directional communication of task-specific settings. For the polishing task, these settings include the meander pattern (e.g., spiral or zig-zag), the number of geometric cycles, the desired polishing force, and the head rotation angle (see Figure 1b). While a sketch is in progress, the wearable also displays a visual progress indicator, providing continuous feedback to the user during the drawing process. Although the system supports a range of wearable devices, we found that smartphones offer advantages such as a larger display and more reliable user interaction.

3.3 Spatial Projection

SketchGuide+ employs a standard areal RGB projector to deliver real-time visual feedback directly onto the workpiece surface. During setup, we perform stereo calibration [15] to align the projector's frame with the RGB-D camera's coordinate system. At runtime, this calibration enables spatially accurate on-surface projections that enhance user interaction. Specifically, when a new sketch begins, an indicator glyph is projected at the sketch's starting point. As the sketch progresses, the projection is dynamically updated to reflect the drawn path in real time (see Figure 1a). These visual cues are essential for providing users with immediate, intuitive feedback and ensuring a seamless sketching experience.

3.4 Motion Planning

In *SketchGuide+*, motion planning is driven by the combination of user-defined sketch trajectories and task-specific parameters provided via the wearable interface. Each sketch consists of a temporally ordered sequence of 3D contact points on the workpiece

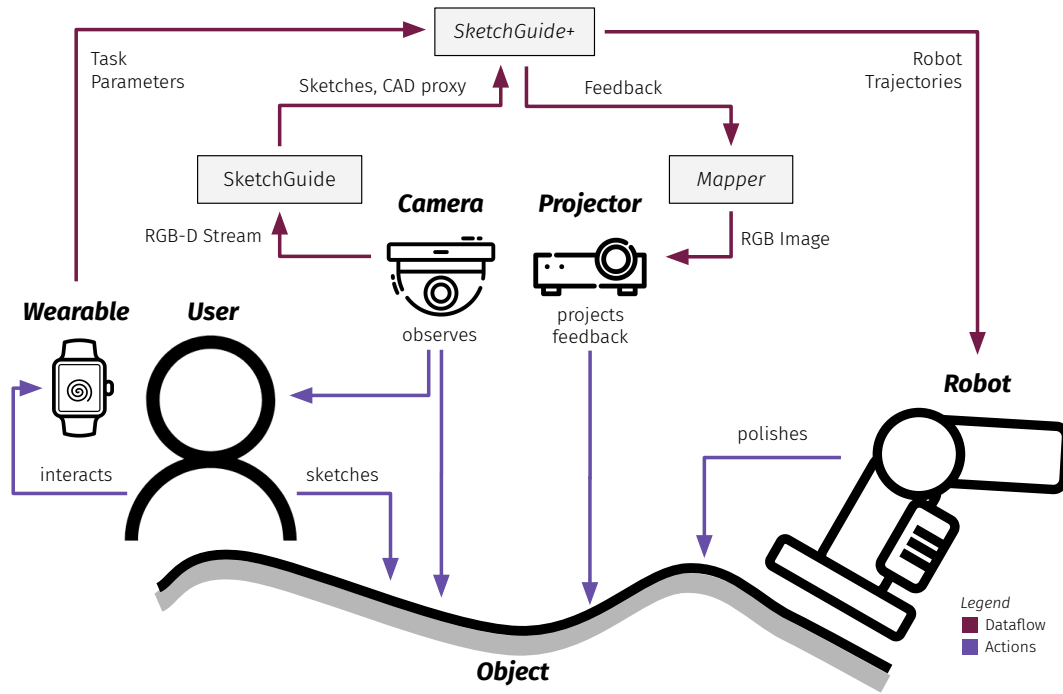


Figure 2: *SketchGuide+* extends the original *SketchGuide* [6] by combining hand-drawn 3D sketches on arbitrary surfaces with task-level parameter input from a smartwatch-style wearable, and by incorporating spatial projection for real-time, on-surface visual feedback. The system operates using a single RGB-D stream to jointly observe both the user and the workpiece. Sketch trajectories are enriched with semantic parameters provided via the wearable interface, and this combined information is used to generate interactive visual feedback and task-aware robot motions.

surface, enriched with local surface normals estimated from fused depth data. These oriented trajectories define the desired motion area in task space. The associated parameters—such as meander pattern, number of cycles, polishing force, and tool orientation—further constrain the trajectory generation process. Based on this input, *SketchGuide+* computes a task frame along the sketch path and uses the estimated surface geometry to maintain appropriate tool-surface alignment. Robot motion is then planned using inverse kinematics to ensure that the end-effector adheres to both the geometric path and the semantic task constraints, while satisfying joint and reachability limits.

4 Evaluation and Results

We evaluate *SketchGuide+* in multiple ways.

4.1 Feasibility Assessment

To assess the feasibility of our approach, we developed an early-stage prototype of *SketchGuide+* and conducted a series of informal tests in a representative polishing scenario. Our evaluation focused on validating the core components of the system pipeline: sketch capture, parameter input via the wearable device, real-time projection feedback, and robot motion execution.

We verified that users could successfully draw freehand sketches on a variety of workpiece surfaces, and that the system reliably

captured and segmented these trajectories in 3D. The smartwatch-style wearable allowed task-specific parameters—such as meander type, number of passes, and tool orientation—to be adjusted in real time, with the progress indicator providing helpful feedback during sketching. Visual feedback also remained clearly perceivable, even on glossy or reflective surfaces. Sketches and their associated parameters were consistently translated into executable robot trajectories.

4.2 User Study

To gain early insights into the usability and perceived benefits of *SketchGuide+*, we conducted a within-subjects study.

4.2.1 Conditions. Our study compares two interface conditions: (1) *SketchGuide+*: our full system with on-surface projection and wearable input, and (2) *SketchGuide*: a baseline without projection feedback and no wearable device using a desktop UI for task-parameter settings and trajectory visualization. The goal was to understand how spatial feedback and integrated parameter control affect user confidence and workflow continuity.

4.2.2 Participants and Tasks. We recruited 6 participants (3 female, 3 male; aged 22–35) with no prior experience in robot programming. Each participant was asked to sketch three polishing paths on a washbasin and specify predefined set of task parameters (e.g.,

Condition	Confidence (M±SD)↑	Ease of Parameters (M±SD)↑	Interaction Flow (M±SD)↑	Completion Time (M±SD)↓ [sec]
<i>SketchGuide+</i>	4.5 ± 0.5	4.3 ± 0.7	4.2 ± 0.9	48.4 ± 3.0
Baseline	3.3 ± 0.9	3.0 ± 1.1	3.3 ± 0.9	52.8 ± 3.6

Table 1: Higher scores indicate better ratings for Confidence in sketch capture, Ease of Parameter input, and perceived Interaction Flow. Completion time is reported in seconds (lower is better). *SketchGuide+* outperforms the baseline across all measures, suggesting improved user experience and efficiency.

meander pattern, number of cycles). All participants completed the task in both conditions, with condition order randomized.

4.2.3 Measures. After both conditions, participants completed a short questionnaire with three Likert-scale (1-5) items (higher is better): (i) How confident were you that your sketch was correctly captured? (ii) How easy was it to adjust task parameters? (iii) How smooth and intuitive did the interaction feel overall? Participants also provided open-ended comments on usability and preferences. Additionally, we tracked the total time to complete each condition.

4.2.4 Results. Participants rated *SketchGuide+* higher than the baseline across all subjective measures (see Table 1.) Confidence in sketch capture improved from 3.3 to 4.5, indicating that the real-time projection enhanced trust in system feedback. Parameter input was also rated easier (4.3 vs. 3.0), suggesting that the wearable interface supported more seamless interaction. Overall interaction flow improved (4.2 vs. 3.3), and average task completion time was slightly reduced (48.4 vs. 52.8), pointing to greater efficiency with the integrated system. Qualitative feedback highlighted that real-time projection made the sketching process feel “clearer” and “more responsive,” while the wearable interface was perceived as “convenient” and “less interruptive” than switching to a desktop. However, one user noted that a noticeable delay in the projection occasionally disrupted their sketching rhythm and made it harder to gauge system responsiveness in fast drawing motions.

4.3 Video

A demonstration video (see Footnote 1) accompanies this work, showcasing the full workflow from user input to robot execution.

5 Discussion

SketchGuide+ explores the integration of freehand sketching, wearable input, and projection-based feedback to enable expressive, low-barrier robot programming on arbitrary physical surfaces. Building on the foundation of *SketchGuide* [6], our system expands the interaction space by fusing geometric and semantic input streams in real time using only a single RGB-D camera, a smartwatch-style wearable, and a standard projector. The goal is to empower users—particularly non-experts—to define robot tasks in situ, without the need for CAD models, tracked tools, or offline programming workflows.

Our preliminary study supports the viability of this approach. Participants rated the full *SketchGuide+* interface higher in terms of confidence, ease of use, and interaction flow compared to a simplified baseline. The addition of on-surface projection was repeatedly

cited as improving spatial awareness and helping users better understand how their input was being interpreted by the system. Likewise, the wearable interface allowed for seamless parameter specification using the same hand involved in sketching, minimizing task interruptions and reducing cognitive load. Together, these components create a workflow that feels responsive, situated, and adaptable—key attributes for flexible manufacturing and human-robot collaboration scenarios.

Nonetheless, several limitations emerged. First, some participants reported a perceptible delay in projection updates, which occasionally disrupted their sketching rhythm. This highlights the need for low-latency feedback, particularly when users draw rapidly or perform corrections. Second, while the wearable was generally well received, interacting with a small screen during active sketching can be imprecise. In practice, users prefer larger wearable displays (e.g., smartphones) over smaller smartwatch-like devices.

Despite these challenges, *SketchGuide+* demonstrates the potential of unifying spatial interaction, semantic parameterization, and real-time visual feedback into a cohesive interface for robot programming. Its reliance on consumer-grade hardware and minimal setup further supports deployment in practical, resource-constrained environments. Our findings contribute to a growing body of work in intuitive, human-centered robot programming and suggest clear pathways for refinement and scalability.

6 Outlook

We are planning a structured usability study to evaluate *SketchGuide+* in comparison to other existing task-level robot programming methods, such as tracked-probe and voice input approaches.

Future extensions will also include support for additional tasks such as inspection, painting, and marking, as well as broader benchmarking across different surface geometries and robotic platforms.²

References

- [1] André Correia and Luís A Alexandre. 2024. A survey of demonstration learning. *Robotics and Autonomous Systems* 182 (2024), 104812.
- [2] Fatih Dogangun, Serdar Bahar, Yigit Yildirim, Bora Toprak Temir, Emre Ugur, and Mustafa Doga Dogan. 2025. RAMP: Robotic Augmented Reality for Machine Programming by Demonstration. *IEEE Robotics and Automation Letters* (2025).
- [3] HC Fang, SK Ong, and AYC Nee. 2013. Orientation planning of robot end-effector using augmented reality. *The International Journal of Advanced Manufacturing Technology* 67 (2013), 2033–2049.
- [4] Anna Fuste, Ben Reynolds, James Hobin, and Valentin Heun. 2020. Kinetic ar: A framework for robotic motion systems in spatial computing. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems*. 1–8.

²Funding: This work was supported by DemoDatenPro (Wi-2022-603642) and the AIT Lighthouse project.

- [5] Soheil Habibian, Antonio Alvarez Valdivia, Laura H. Blumenschein, and Dylan P. Losey. 2024. A survey of communicating robot learning during human-robot interaction. *The International Journal of Robotics Research* (2024).
- [6] Christoph Heindl, Gerhard Ebenhofer, Sezgin Kutlu, Fabian Widmoser, and Andreas Pichler. 2025. SketchGuide: A Baseline Vision-Based Model for Rapid Robot Programming via Freehand Sketching on Any Surface. In *European Robotics Forum 2025*, Marco Huber, Alexander Verl, and Werner Kraus (Eds.). Springer Nature Switzerland, Cham, 174–179.
- [7] Wojciech Kaczmarek, Jaroslaw Panasiuk, Szymon Borys, and Patryk Banach. 2020. Industrial robot control by means of gestures and voice commands in off-line and on-line mode. *Sensors* 20, 21 (2020), 6358.
- [8] Junsoo Kim, Sangyeop Lee, and Joonbum Bae. 2025. An augmented reality-based wearable system for handheld-free and intuitive robot programming. *Journal of Intelligent Manufacturing* (2025), 1–10.
- [9] Guoliang Liu, Wenlei Sun, and Pinwen Li. 2024. Motion capture and AR based programming by demonstration for industrial robots using handheld teaching device. *Scientific Reports* 14, 1 (2024), 23259.
- [10] Sina Masnadi, Joseph J LaViola Jr, Jana Pavlasek, Xiaofan Zhu, Karthik Desingh, and Odest Chadwicke Jenkins. 2019. Sketching affordances for human-in-the-loop robotic manipulation tasks. *2nd Robot Teammates Operating in Dynamic, Unstructured Environments (RT-DUNE)* (2019).
- [11] David Porfirio, Laura Stegner, Maya Cakmak, Allison Sauppé, Aws Albarghouthi, and Bilge Mutlu. 2023. Sketching robot programs on the fly. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. 584–593.
- [12] Jaison Puthenkalam, Setareh Zafari, Andreas Sackl, Katja Gallhuber, Gerhard Ebenhofer, Markus Ikeda, and Manfred Tscheligi. 2023. Where should I put my Mark? VR-based Evaluation of HRI Modalities for Industrial Assistance Systems for Spot Repair. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication*.
- [13] Hang Su, Wen Qi, Jiahao Chen, Chenguang Yang, Juan Sandoval, and Med Amine Laribi. 2023. Recent advancements in multimodal human-robot interaction. *Frontiers in Neurobotics* 17 (2023).
- [14] Digang Sun, Ping Zhang, and Mingxuan Chen. 2023. A versatile interaction framework for robot programming based on hand gestures and poses. *Robotics and Computer-Integrated Manufacturing* 84 (2023), 102596.
- [15] Zhengyou Zhang. 2002. A flexible new technique for camera calibration. *IEEE Transactions on pattern analysis and machine intelligence* 22, 11 (2002), 1330–1334.